

Introduction

The bootstrap offers several ways to construct confidence intervals from resamples. They differ in accuracy (first- vs second-order correctness), in assumptions, and in complexity. Choosing well among them is part of responsible bootstrap use.

Prerequisites

Bootstrap, confidence intervals.

Theory

For a statistic $\hat{\theta}$ with bootstrap replicates $\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$:

- **Percentile:** $[\hat{\theta}_{(\alpha/2)}^*, \hat{\theta}_{(1-\alpha/2)}^*]$. Simple and intuitive; assumes symmetry of the bootstrap distribution.
- **Basic (pivotal):** $[2\hat{\theta} - \hat{\theta}_{(1-\alpha/2)}^*, 2\hat{\theta} - \hat{\theta}_{(\alpha/2)}^*]$. Uses the bootstrap distribution of $\hat{\theta}^* - \hat{\theta}$ as a pivot.
- **Studentised (bootstrap-t):** requires bootstrap SE, generally most accurate but computationally heavier.
- **BCa (bias-corrected and accelerated):** corrects for bias and skewness; the default modern choice.

Coverage: percentile is first-order accurate (error $O(n^{-1/2})$); BCa and studentised are second-order ($O(n^{-1})$).

Assumptions

Same as bootstrap generally; specific methods may assume additional smoothness.

R Implementation

```
library(boot)
set.seed(2026)

x <- rlnorm(40, meanlog = 0, sdlog = 0.8) # skewed data

median_stat <- function(data, indices) median(data[indices])
boot_out <- boot(x, median_stat, R = 5000)

ci <- boot.ci(boot_out, type = c("norm", "basic", "perc", "bca"))
ci

# Studentised bootstrap
median_with_var <- function(data, indices) {
  d <- data[indices]
  md <- median(d)
  # Jackknife variance of median
  jk <- sapply(seq_along(d), function(i) median(d[-i]))
  c(md, var(jk) * (length(d) - 1))
}
boot_t <- boot(x, median_with_var, R = 5000)
boot.ci(boot_t, type = "stud")
```

Output & Results

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS

Normal (0.728, 1.241)

Basic	(0.712, 1.221)
Percentile	(0.735, 1.244)
BCa	(0.800, 1.297)

Studentized (0.729, 1.341)

For skewed data, BCa shifts the interval to the right, correcting for the bias in the percentile CI.

Interpretation

“The median was 0.97 with a 95 % BCa bootstrap CI of (0.80, 1.30) based on 5000 resamples.”

Practical Tips

- Default to BCa CIs unless computational constraints matter; they correct for bias and skew at modest additional cost.
- For very small B (e.g., $B < 1000$), percentile and BCa can be unstable at the tails; increase B .
- Studentised (boot-t) CIs are the most accurate but require a variance estimate of the statistic (jackknife or formula); worth it for critical inference.
- Normal bootstrap CIs (using mean ± 1.96 SE) are the crudest; avoid them unless the bootstrap distribution is nearly normal.
- Report which CI method was used and the number of bootstrap resamples.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests